

Sandunika Ranasinghe

Department of Electrical and Electronic Engineering, University of Peradeniya

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RESEARCH INTERESTS

I am interested in leveraging **deep learning, image and signal processing**, and **applied mathematics** to develop theoretically grounded models with strong real-world applicability.

EDUCATION

BSc. (Hons.) in Electrical and Electronic Engineering (First Class Honours) University of Peradeniya, Sri Lanka	2021 March- 2025 August GPA : 3.95/4.00 Class Rank : 4/115
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G.C.E. Advance level examination Physics (A), Chemistry (A), Combined Mathematics (A)	2019 National Rank - 167/32000+
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PUBLICATIONS

J – JOURNALS, C – CONFERENCES

- J 1 Reflectance Multispectral Imaging for Soil Texture Composition Estimation and USDA Texture Classification**
in *IEEE Access* [Under review] [DOI](#)
S. Ranasinghe, S. Jayakody, M. De Silva, G. Thilakarathne, R. Godaliyadda, V. Herath, M. P. Ekanayake, S. Navaratnarajah.
- C 1 Feature Conditioned Diffusion for Audio Generation**
[DOI](#)
in *19th International Conference on Industrial and Information Systems 2025 (ICIIS 2025)*
S. Perera, **S. Ranasinghe**, S. Jayakody, B. Epakanda, R. Godaliyadda, M. P. Ekanayake.
- C 2 Multispectral Imaging and Machine Learning for Soil Classification through Moisture-Induced Spectral Dynamics**
[DOI](#)
in *19th International Conference on Industrial and Information Systems 2025 (ICIIS 2025)*
G. Thilakarathne, S. Perera, **S. Ranasinghe**, S. Jayakody, R. Godaliyadda, V. Herath, M. P. Ekanayake, S. Navaratnarajah.
- C 3 Deep Learning for Soil Moisture Content Estimation via Reflectance Multispectral Imaging**
[DOI](#)
in *9th International Conference on Advancements in Technology and Computing 2024 (ICATC 2024)*
S. Ranasinghe, S. Jayakody, M. De Silva, V. Herath, R. Godaliyadda, P. Ekanayake, S. Navaratnarajah, F. Kizel, G. Thilakarathne.

HIGHLIGHTED ACHIEVEMENTS

- Awarded **second runners up** in the **Best Undergraduate Project in Sri Lanka - Manamperi Award 2025** for the Final Year Project on "Multispectral Imaging"
- Awarded the **Silver Award** in the Engineering field at **Sahasak Nimawum Inventions and Innovations Competition 2024** for the Final Year Project on "Multispectral Imaging"
- Second runners up** in the Healthcare category at **ACES Hackathon 2023**, organized by the Association of Computer Engineering Students, University of Peradeniya.
- Finalists** at the **"Predicta v1.0" 2023**, organized by IEEE Student Branch of University of Peradeniya.
- Finalists** at the **"IEEE Electronic Design Competition" 2023**, Organized by IEEE Sri Lanka Section.
- Received **High Distinction Pass** at the **"Australian National Chemistry Quiz (Senior Division)" 2018**.

SELECTED PROJECTS

HeritageAI

2025 - Present

Under Nvidia Academic Grant Program

Focused on developing a unified restoration pipeline for degraded Sri Lankan historical images using implicit neural representations (INRs) and diffusion models, with current work focused on INR based modules for denoising and Diffusion based models for colorization

Concepts: Diffusion Models, Implicit Neural Representations (INRs), Image Restoration.

Denoising Diffusion Models

2025 - Present

Conducted research on denoising diffusion models, focusing on improving generative fidelity and extending diffusion frameworks to structured audio signal generation.

Key studies:

- **Improving Diffusion Probabilistic Models (2025 - Present):** Incorporated isotropy-inspired constraints into the diffusion training objective to address distributional limitations in standard DDPMs. Achieved measurable improvements in generative fidelity across evaluation metrics.
- **Diffusion for Audio Signals (2025):** Investigated temporal structure representations for diffusion-based audio generation. Conditioned models on signal-processing-based temporal features to enhance waveform stability, temporal coherence, and perceptual realism relative to DiffWave.

Concepts: Denoising Diffusion Probabilistic Models (DDPM), Generative Modeling, Isotropy Constraints, Audio Signal Processing, Temporal Representation Learning, DiffWave.

Multispectral Imaging for Agricultural and Environmental Monitoring

2023 - 2025

Conducted multispectral sensing research for environmental and industrial analysis, developing machine learning models for soil characterization and algal bloom detection.

Key studies:

- **Soil Moisture Content Estimation (2023 - 2024):** Built regression models to estimate moisture levels from spectral features.
- **Soil Classification and Composition Prediction (2024 - 2025):** Developed machine learning models to classify soil texture and predict composition.
- **Microcystis Cell Count and Toxicity Estimation (2025):** Designed regression models to detect harmful algal blooms.

Concepts: Multispectral Sensing, Feature Extraction, Regression and Classification Models, Neural Networks.

Survey Buddy

2024

Developed a surveying device using easily accessible and cost effective modules along with sophisticated sensor fusion algorithms to map terrain.

Concepts: Kalman-filter, Signal Processing, Mobile Terrain Mapping

GloSmart

2024

Developed a cyclist safety system that involved programming in Assembly language.

Concepts: Embedded Systems, Microcontroller Programming, Assembly Language

Non-Invasive Blood Glucose Sensor

2023

Developed a blood glucose sensor to give a high-accuracy reading of glucose concentration without skin penetration.

Concepts: Near-Infrared and Mid-Infrared Spectroscopy, Signal Processing

EXPERIENCE

Research and Development Engineer

2025 November - present

Farbe Technologies

Developing AI-driven signal and image processing solutions for industrial applications.

Volunteer Researcher

2023 June - present

Multidisciplinary AI Research Center, University of Peradeniya

Served as a volunteer researcher at MARC, contributing to the following theoretical and applied machine learning and deep learning projects across two research groups:

- Diffusion-based Generative AI
- Multispectral Imaging for agricultural and environmental monitoring

Teaching Assistant

2025 August - 2025 October

Department of Electrical and Electronic Engineering, University of Peradeniya

Provided guidance to students in EE358 power module and EE1010 electricity module.

Trainee Research and Development Engineer

2023 July – 2023 October

SLT Digital Lab, Sri Lanka Telecom

Contributed to the 'Deep Learning for RACOON AI' project, working on object detection and tracking using YOLOv8, deploying machine learning models with Seldon Core, managing databases with PostgreSQL, and utilizing ONNX for model interoperability.

SERVICES

Public Speaker/Volunteer - AI Workshops

2025

Participated in workshops to share knowledge on generative AI and Multispectral Imaging for school students through the Multidisciplinary AI Research Center at the University of Peradeniya.

Executive committee member, IET on Campus, University of Peradeniya

2022 June - 2023 May

Assisted in organizing technical events and workshops to undergraduates.

Reporting team member, IEEE Robotics and Automation Society, University of Peradeniya

2022

Supported the coordination of technical events and workshops to undergraduates, while also contributing to editorial content and documentation.

Team member, OGV/H4TF CXP, AIESEC in Kandy

2022

Engaged in tasks related to tracking customer experience in AIESEC projects.

Executive committee member, Astronomical Society of Mahamaya Girls' College, Kandy

2018-2019

Organized astronomical nights and workshops

SKILLS

- **Programming Languages** - Python, C, Assembly
- **Libraries frameworks** - TensorFlow, PyTorch, Scikit- Learn, Keras, ONNX, YOLOv8, cv2, Scipy
- **Research** - Academic research, Project documentation, Technical writing
- **Other** – Latex, Git, SQL, PostgreSQL

REFERENCES

Prof. Roshan Godaliyadda

Ph.D.(National University of Singapore)

SMIEEE, AMIESL

Professor

Department of Electrical and Electronic Engineering
Faculty of Engineering, University of Peradeniya, Sri Lanka

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Prof. H.M.V.R. Herath

Ph.D.(University of Miami, USA)

SMIEEE, AMIESL

Professor

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